

Mapping Nigeria's Place in the Global Prostate Cancer Knowledge Graph

A Computational Wikidata Network Analysis

Concept Note Presentation

Focus: prostate cancer · Wikidata · Nigeria · representation audit

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Background and Rationale

Why this study matters

Public-health concern

Prostate cancer remains a leading cancer burden among men, with late diagnosis and limited awareness worsening outcomes.

Knowledge ecosystem

Open knowledge platforms now shape how disease information is structured, searched, reused, and represented.

Africa gap

Wikimedia studies show geographic imbalances in participation and content, with African countries under-represented.

Core premise: representation gaps in structured knowledge graphs can affect what becomes visible, reusable, and editable about African health realities.



Research Gap

What has not yet been adequately examined

Known

- Prior Wikimedia research has documented geographic bias and unequal participation.
- Most attention has been on Wikipedia article-level content.

Unresolved

- Whether similar imbalance exists inside Wikidata's machine-readable entity networks.
- Where Nigeria appears or disappears within the prostate cancer knowledge graph.

Gap statement: No disease-specific computational mapping has yet clarified Nigeria's representation in the Wikidata prostate cancer graph.



Problem Statement

Two issues motivate the study

1. Representation uncertainty

The number of clinical trials and publications linked to prostate cancer appears small relative to the disease's global research footprint.

2. Data-quality concern

Some SPARQL outputs appear mismatched to unrelated items, such as works of fiction or films sharing labels with publications.

Therefore, the study must audit both representation and data quality before recommending Wikidata improvements.



Aim and Objectives

Precise direction of the study

Aim

To computationally map the structure of the prostate cancer knowledge graph on Wikidata and quantify Nigeria's representation within it.

Objectives

01

Construct and analyze the prostate cancer entity network on Wikidata.

02

Quantify Nigerian representation in linked trials and publications, then benchmark against reference disease figures.

03

Generate prioritized recommendations for targeted Wikidata contributions.



Data Foundation

Initial Wikidata extracts and planned expansion

Two-hop network

1,124 relations rooted at the prostate cancer item.

Treatment set

99 relations covering genetic associations, therapies, and symptoms.

Trials list

Clinical trials linked to the prostate cancer entity.

Publications list

Linked publication entities requiring representation and quality checks.

Additional SPARQL

Further querying will expand entities, properties, and relations.

Versioning

Each data pull will be dated and preserved for reproducibility.



Methodological Workflow

From Wikidata extraction to actionable recommendations



Final output: prioritized Wikidata items and statements to add, correct, or expand for stronger Nigeria-focused representation.



Network Analysis Plan

How the Wikidata graph will be examined

Graph model

Directed multigraph computed in R programming language.

Entity classes

Genes, drugs, symptoms, conditions, and taxa typed by connecting property semantics.

Centrality

Degree and eigenvector centrality will identify structurally important entities.

Communities

Louvain community detection will reveal clusters such as genetic-association and treatment groups.

Interpretation

Network position will guide which areas require correction, expansion, or Nigeria-focused enrichment.



Representation and Benchmarking Audit

Testing whether Nigeria is visible in linked evidence

Audit question

Are Nigerian clinical trials, publications, institutions, authors, or disease-burden realities visible within the Wikidata prostate cancer graph?

Wikidata-linked evidence
Clinical trials and publications associated with the prostate cancer item.

Benchmark sources
GLOBOCAN, Nigerian cancer-registry sources, WHO, and Ministry of Health figures.

Audit outcome
A quantified account of under-representation, over-representation, or data-quality gaps.



Expected Results

What the study will deliver

- A mapped prostate cancer Wikidata network showing central entities and communities.
- A benchmarked audit of Nigeria's representation in linked trials and publications.
- An entity-quality audit identifying mismatches and problematic records.
- A prioritized list of Wikidata items and statements to add, correct, or expand.
- A scientific manuscript targeted at a reputable Scopus-indexed journal.

Value: the work moves Wikidata improvement from general awareness to evidence-based, targeted intervention.



Conclusion

This study will show where Nigeria is visible, missing, or misrepresented in the global prostate cancer knowledge graph—and translate those findings into practical Wikidata improvement priorities.

Computational

Evidence-based

Nigeria-focused

Wikidata-ready



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